

Sudar Manikandan S – Thiagarajar College of Engineering

Structured Training Program: Networking

Week 3

Module 5: Layer 3 Session 1

1. Analyse the ARP packet using Wireshark. Inspect the ARP request and reply frames, and discuss the role of the sender's IP and MAC address in these packets.

a) Captured an filtered only the ARP packet in the Wireshark.

The screenshot shows the Wireshark interface with the filter 'arp' applied. The packet list displays 112 ARP packets. The packet details pane shows the selected packet (No. 1) as an ARP request (opcode 1) from Sophos_cf:f2:a5 to the broadcast address ff:ff:ff:ff:ff:ff. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Sophos_cf:f2:a5	Broadcast	ARP	60	Who has 10.13.244.62? Tell 10.13.128.1
3	0.005338	Sophos_cf:f2:a5	Broadcast	ARP	60	Who has 10.13.239.26? Tell 10.13.128.1
9	0.005338	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.255.246
13	0.102169	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.255.2
16	0.106418	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.245.154
34	0.309967	Sophos_cf:f2:a5	Broadcast	ARP	60	Who has 10.13.233.219? Tell 10.13.128.1
39	0.412929	RuijieNetwork...	Broadcast	ARP	60	Who has 10.11.128.1? Tell 10.11.128.11
56	0.623545	DLinkInterna...	Broadcast	ARP	60	Who has 172.17.44.62? Tell 172.17.44.61
57	0.623545	DLinkInterna...	Broadcast	ARP	60	Who has 172.17.44.62? Tell 172.17.44.61
58	0.623545	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.245.145
61	0.623545	RuijieNetwork...	Broadcast	ARP	60	Who has 10.11.128.1? Tell 10.11.128.12
66	0.721067	Intel_ba:d8:6c	Broadcast	ARP	60	Who has 10.13.244.147? Tell 10.13.243.210
67	0.721067	RuijieNetwork...	Broadcast	ARP	60	Who has 10.11.128.1? Tell 10.11.196.7
68	0.729832	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.245.147
69	0.750333	AzureWaveTec...	Broadcast	ARP	60	Who has 10.13.245.75? (ARP Probe)
74	0.827498	MicrochipTec...	Broadcast	ARP	60	Who has 172.18.64.1? Tell 172.18.64.6
77	0.827498	Intel_e9:1d:3a	Broadcast	ARP	60	Who has 10.13.243.75? Tell 10.13.244.155
80	0.827498	Sophos_cf:f2:a5	Broadcast	ARP	60	Who has 10.13.240.232? Tell 10.13.128.1
82	0.828394	6e:e0:82:2c:6...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.241.116
84	0.928948	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.255.248
88	1.127526	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.245.156
98	1.437306	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.255.1
100	1.437306	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.245.160
101	1.437306	RuijieNetwork...	Broadcast	ARP	60	Who has 10.11.128.1? Tell 10.11.128.11
103	1.437306	MicrochipTec...	Broadcast	ARP	60	Who has 172.18.64.1? Tell 172.18.64.6
105	1.437306	3a:f1:72:da:8...	Broadcast	ARP	60	ARP Announcement for 10.13.239.41[Malformed Packet]
106	1.556250	Sophos_cf:f2:a5	Broadcast	ARP	60	Who has 10.13.250.255? Tell 10.13.128.1
107	1.556250	Sophos_cf:f2:a5	Broadcast	ARP	60	Who has 10.13.241.169? Tell 10.13.128.1
108	1.556250	RuijieNetwork...	Broadcast	ARP	60	Who has 10.13.128.1? Tell 10.13.245.152
112	1.650952	DLinkInterna...	Broadcast	ARP	60	Who has 172.17.44.62? Tell 172.17.44.61

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface Ethernet II, Src: Sophos_cf:f2:a5 (7c:5a:1c:cf:f2:a5), Dst: Broadcast (ff:ff:ff:ff:ff:ff) Address Resolution Protocol (request)

b) To inspect only the arp request packet we need to use the filter **arp.opcode == 1**.

The screenshot shows the Wireshark interface with the filter 'arp.opcode == 1' applied. The packet list displays 112 ARP packets. The packet details pane shows the selected packet (No. 1) as an ARP request (opcode 1) from Sophos_cf:f2:a5 to the broadcast address ff:ff:ff:ff:ff:ff. The packet bytes pane shows the raw data in hexadecimal and ASCII.

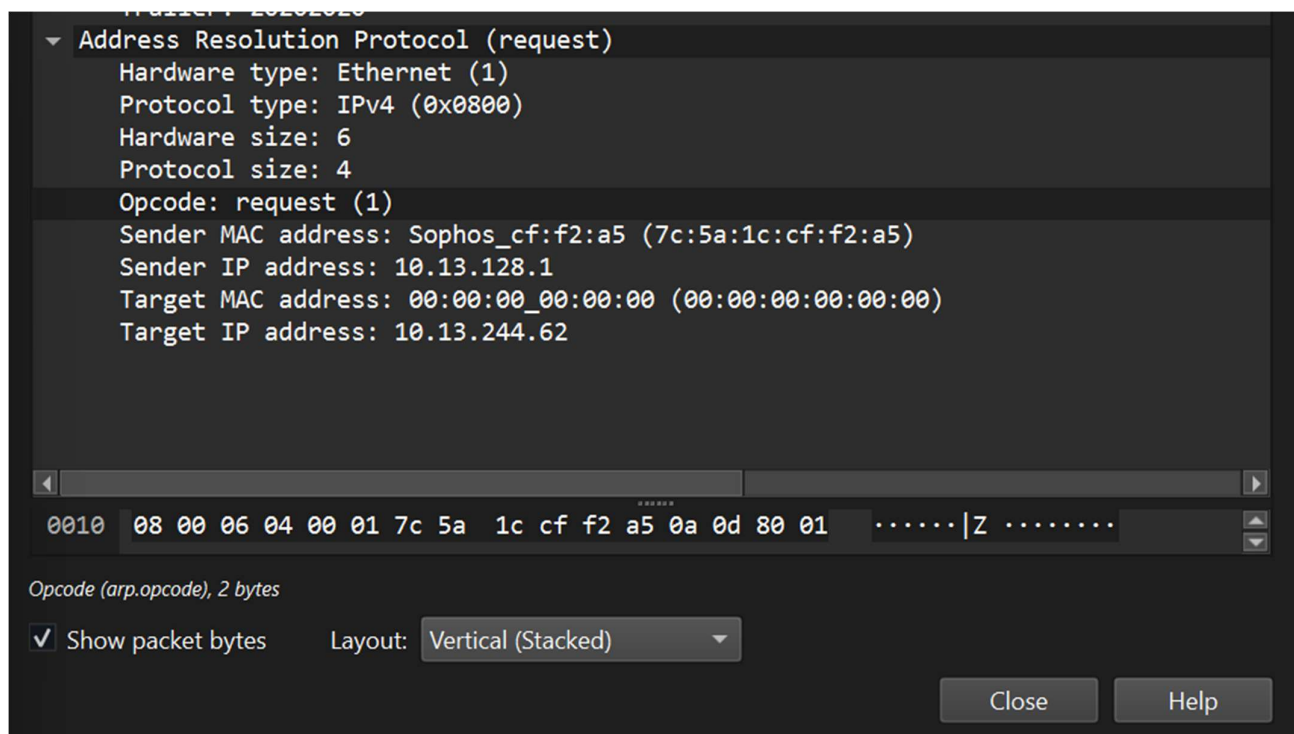
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61	0.623545	RuijieNetwork...	Broadcast	ARP	60	Who has 10.11.128.1? Tell 10.11.128.12
66	0.721067	Intel_ba:d8:6c	Broadcast	ARP	60	Who has 10.13.244.147? Tell 10.13.243.210
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c) Inspecting the request packet I inferred the following things:

The packet consist of the following fields

- **Hardware Type (2 bytes):** Specifies the type of data link layer technology used. It allows ARP to operate across different network technologies. For Ethernet networks, this value is 1.
- **Protocol Type (2 bytes):** Identifies the network protocol whose address is being mapped to a MAC address. It is commonly IPv4 (0x0800).
- **Hardware Size (1 byte):** Specifies the length of the hardware address. For Ethernet, the MAC address length is 6 bytes (48 bits).
- **Protocol Size (1 byte):** Specifies the length of the protocol address. For IPv4, the IP address length is 4 bytes (32 bits).
- **Opcode (2 bytes):** Indicates the type of ARP message.
1 – ARP Request
2 – ARP Reply
- **Sender MAC Address (6 bytes):** Specifies the MAC address of the sender (the device that generated the ARP request).
- **Sender IP Address (4 bytes):** Specifies the IP address of the sender. This allows the receiving device to know where to send the ARP reply and helps the receiver device to update its ARP cache if needed.
- **Target MAC Address (6 bytes):** In an ARP request, this field is typically set to 00:00:00:00:00:00 because the sender does not yet know the MAC address of the target device.
- **Target IP Address (4 bytes):** Specifies the IP address of the device whose MAC address is being requested. The ARP request is broadcast to all devices in the local network to find this address.



d) To filter the arp reply packets we need to set **arp.opcode == 2**

The screenshot shows the Wireshark interface with the filter `arp.opcode == 2` applied. The packet list shows several ARP packets, with packet 1184 highlighted. The packet details pane for packet 1184 shows the following structure:

- Section number: 1
- Interface id: 0 (\Device\NPF_{98D6EAAA-E2D3-471E-A6A0-69804EE724E2})
- Encapsulation type: Ethernet (1)
- Arrival Time: Mar 14, 2026 22:52:48.849617000 India Standard Time
- UTC Arrival Time: Mar 14, 2026 17:22:48.849617000 UTC
- Epoch Arrival Time: 1773508968.849617000
- [Time shift for this packet: 0.000000000 seconds]
- [Time delta from previous captured frame: 2.374300 milliseconds]
- [Time delta from previous displayed frame: 602.480700 milliseconds]
- [Time since reference or first frame: 13.207321500 seconds]
- Frame Number: 1184
- Frame Length: 46 bytes (368 bits)
- Capture Length: 46 bytes (368 bits)
- [Frame is marked: False]
- [Frame is ignored: False]
- [Protocols in frame: eth:ethertype:vlan:ethertype:arp]
- Character encoding: ASCII (0)
- [Coloring Rule Name: ARP]
- [Coloring Rule String: arp]
- Ethernet II, Src: 72:35:d2:ef:e1:4c (72:35:d2:ef:e1:4c), Dst: Intel_55:61:7b (4c:49:6c:55:61:7b)
 - Destination: Intel_55:61:7b (4c:49:6c:55:61:7b)
 - Source: 72:35:d2:ef:e1:4c (72:35:d2:ef:e1:4c)
 - Type: 802.1Q Virtual LAN (0x8100)
 - [Stream index: 215]
- 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
 - 000. = Priority: Best Effort (default) (0)
 - ...0 = DEI: Ineligible
 - 0000 0000 0000 = ID: 0
 - Type: ARP (0x0806)
- Address Resolution Protocol (reply)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: reply (2)
 - Sender MAC address: 72:35:d2:ef:e1:4c (72:35:d2:ef:e1:4c)
 - Sender IP address: 10.13.237.23
 - Target MAC address: Intel_55:61:7b (4c:49:6c:55:61:7b)

e) Inspecting the reply packet, I inferred that:

The packet structure of the ARP reply is similar to that of the ARP request. However, the main difference is that in the reply packet, the Target MAC Address field contains the actual MAC address of the requesting system, instead of being set to 00:00:00:00:00:00 as in the request packet. Additionally, the Sender MAC Address field contains the MAC address of the device whose IP address was requested, indicating the mapping between the IP address and its corresponding MAC address. The ARP reply is then sent as a unicast packet directly to the requesting system.

The screenshot shows the packet details for the ARP reply packet (packet 1184). The structure is as follows:

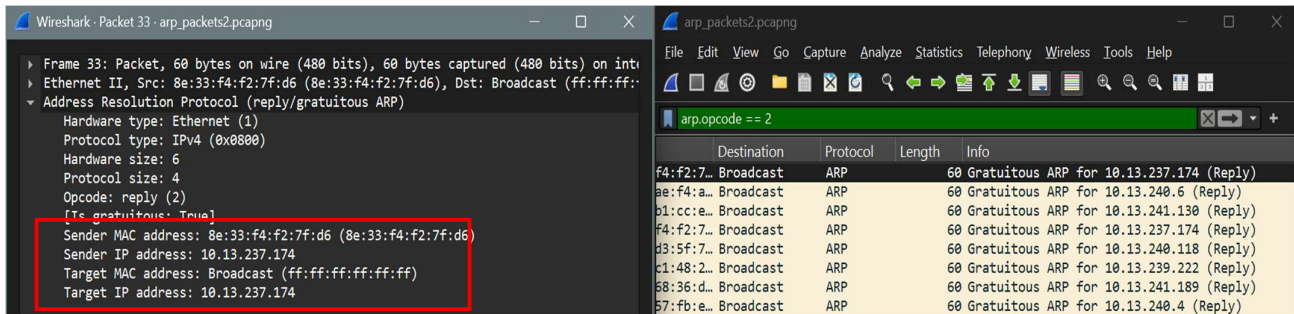
- 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0
- Address Resolution Protocol (reply)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: reply (2)
 - Sender MAC address: 72:35:d2:ef:e1:4c (72:35:d2:ef:e1:4c)
 - Sender IP address: 10.13.237.23
 - Target MAC address: Intel_55:61:7b (4c:49:6c:55:61:7b)
 - Target IP address: 10.13.248.207

The following was the corresponding request packet of the response packet analysed

The screenshot shows the Wireshark interface with the filter `arp.src.proto_ipv4 == 10.13.248.207 || arp.dst.proto_ipv4 == 10.13.237.23` applied. The packet list shows two ARP packets:

- Packet 1158: Time 12.89122, Source Intel_55:61:7b, Destination 72:35:d2:ef:e1:4c, Protocol ARP, Length 42, Info Who has 10.13.237.23? Tell 10.13.248.207
- Packet 1184: Time 13.20732, Source 72:35:d2:ef:e1:4c, Destination Intel_55:61:7b, Protocol ARP, Length 46, Info 10.13.237.23 is at 72:35:d2:ef:e1:4c

Apart from these 2 types of ARP packet there is another kind of ARP reply packet called **Gratuitous ARP**. This ARP reply packets are broadcasted across the network to notify that their IP address has been changed or updated helping the systems in network to update their ARP cache table. This packet contains the src and dst IP as same.



2) Using Packet Tracer, simulate an ARP spoofing attack. Analyze the behaviour of device on the network when they receive a malicious ARP response.

ARP Spoofing Attack:

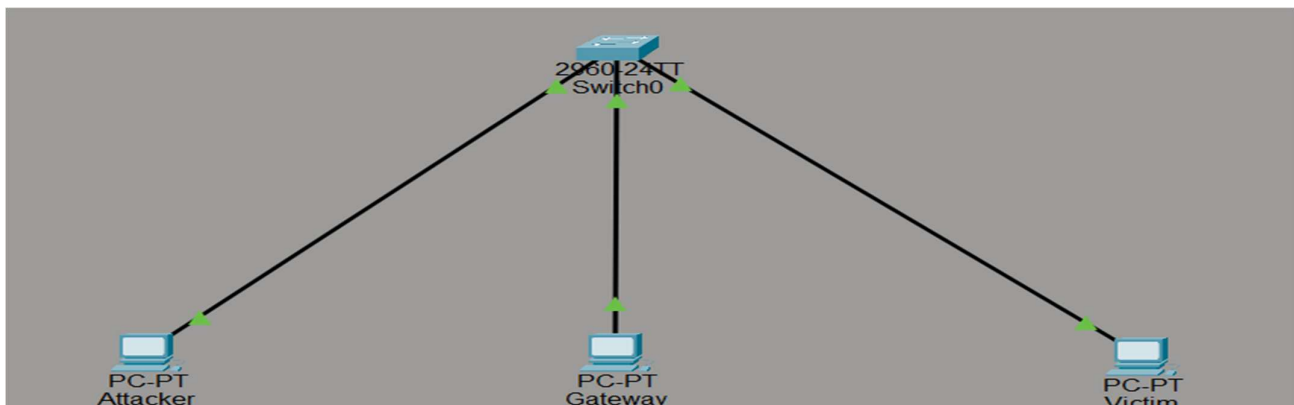
An ARP spoofing attack (or ARP poisoning) is a type of man-in-the-middle (MITM) attack where an attacker sends falsified Address Resolution Protocol (ARP) messages over a local area network (LAN). This associates the attacker's MAC address with the IP address of a legitimate device, such as the default gateway, allowing them to intercept, modify, or stop traffic. To do this in a packet tracer we need

- Attacker Device
- Gateway
- Victim device

Steps:

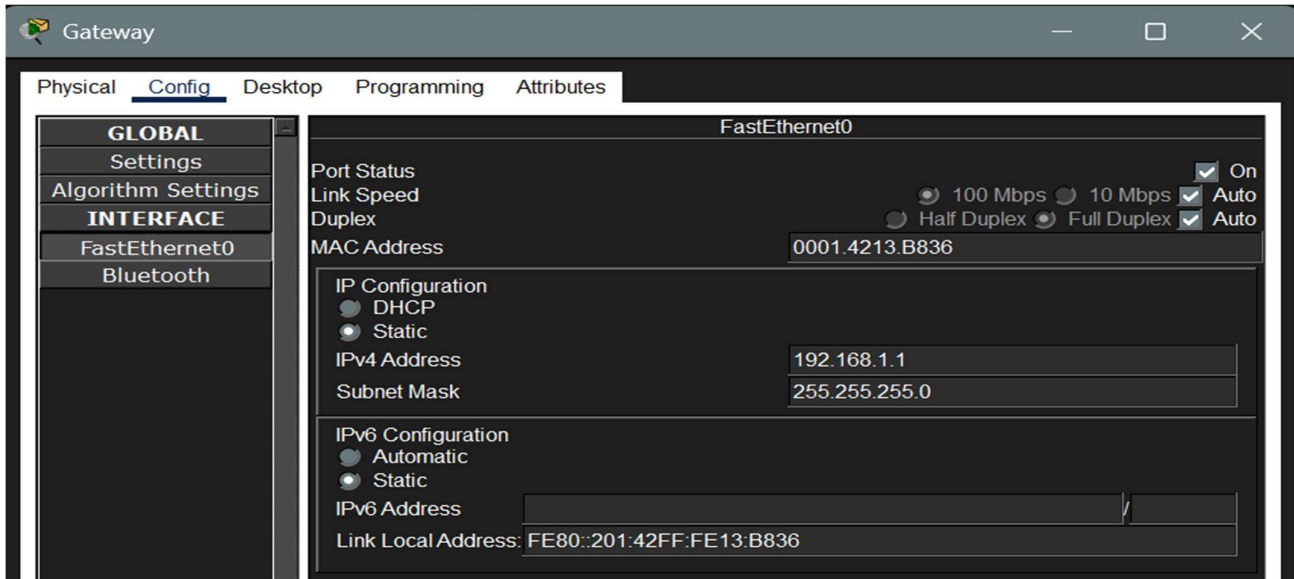
a. Topology Setup

- Place a Switch.
- Add three PCs: Victim, Attacker, and Gateway (a PC or a router)
- Connect all devices to the switch using cables.

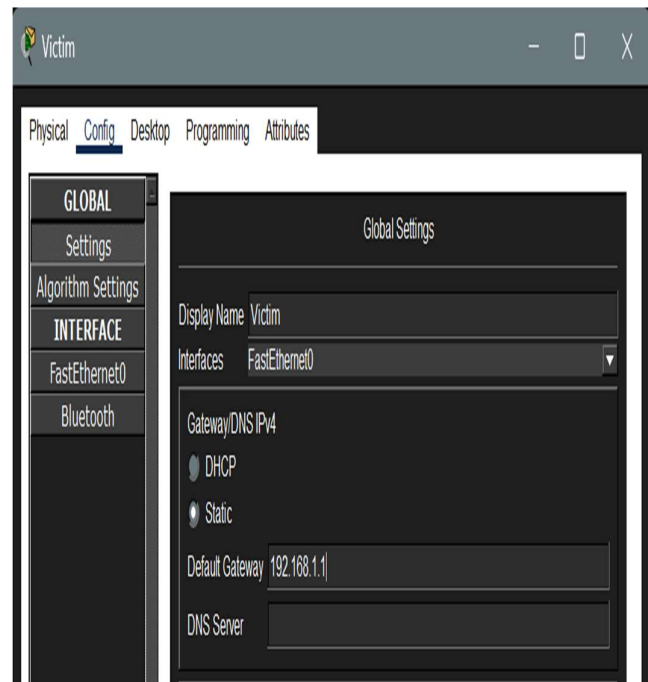
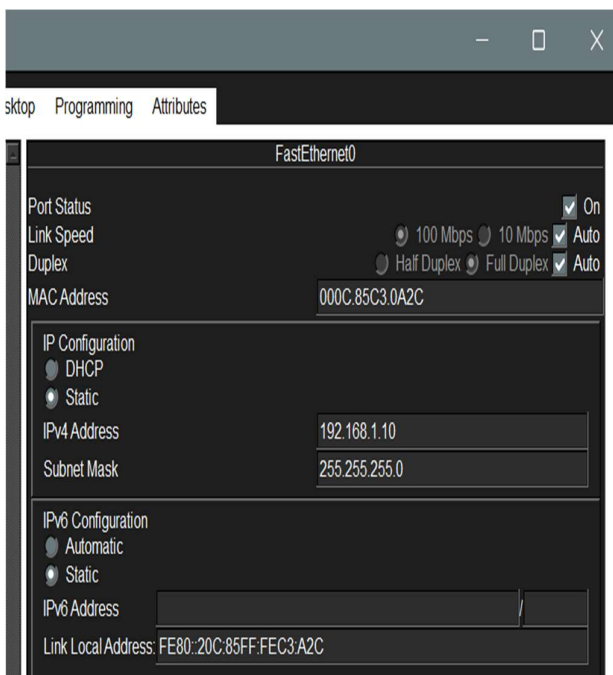


b. IP Configuration

- Gateway
 - IP Address: 192.168.1.1
 - Subnet Mask: 255.255.255.0
 - MAC Address: 00D0.D3EB.1B44

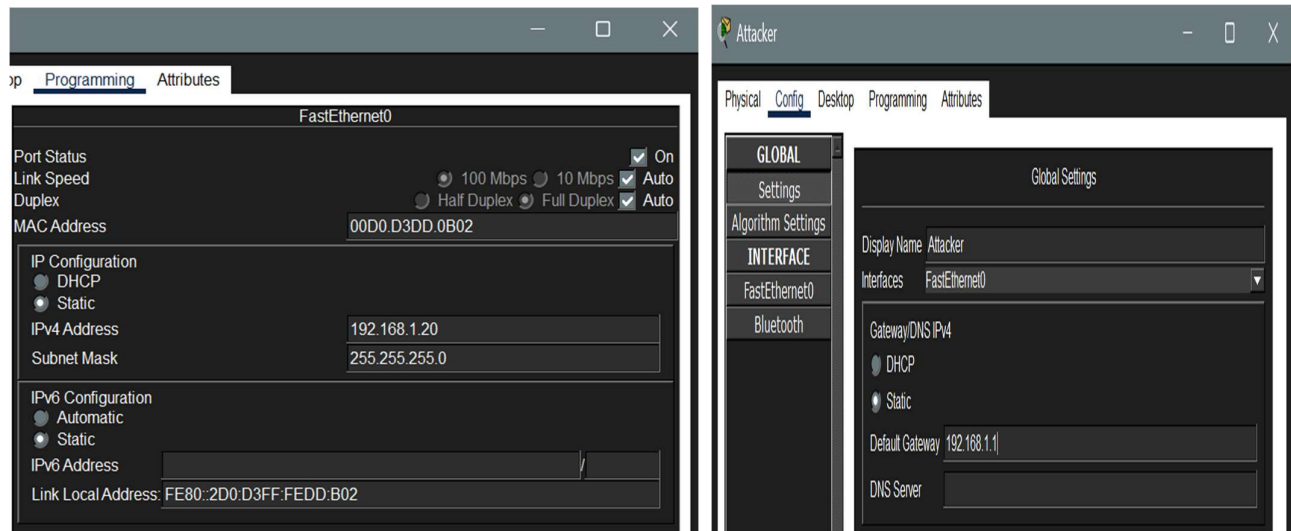


- Victim
 - IP Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1



Attacker

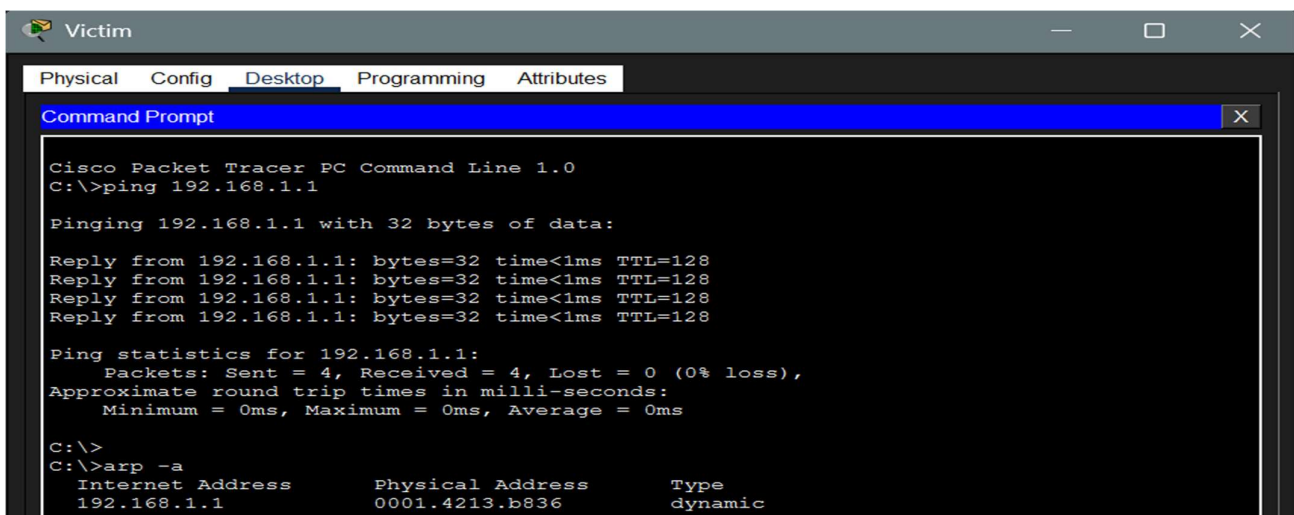
- IP Address: 192.168.1.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1



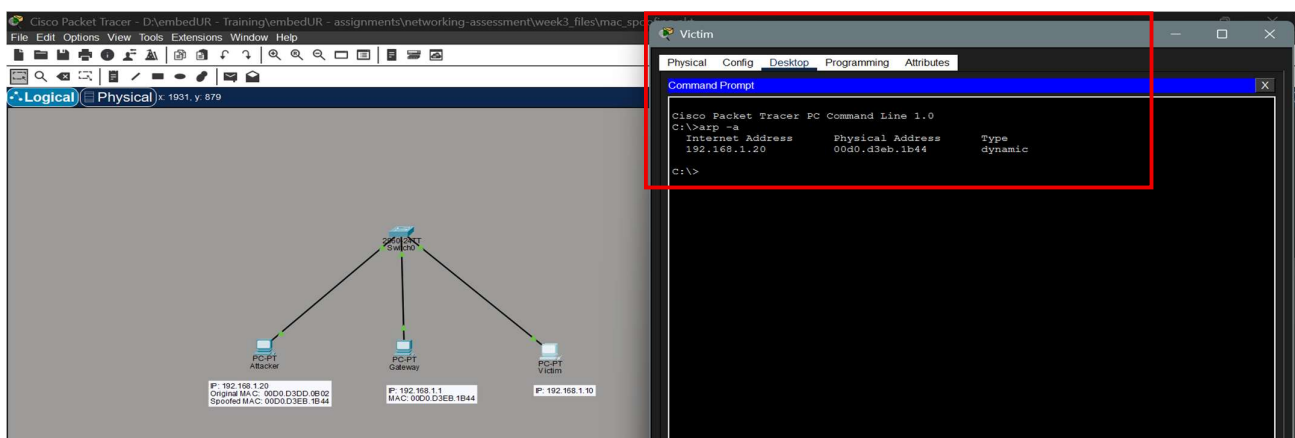
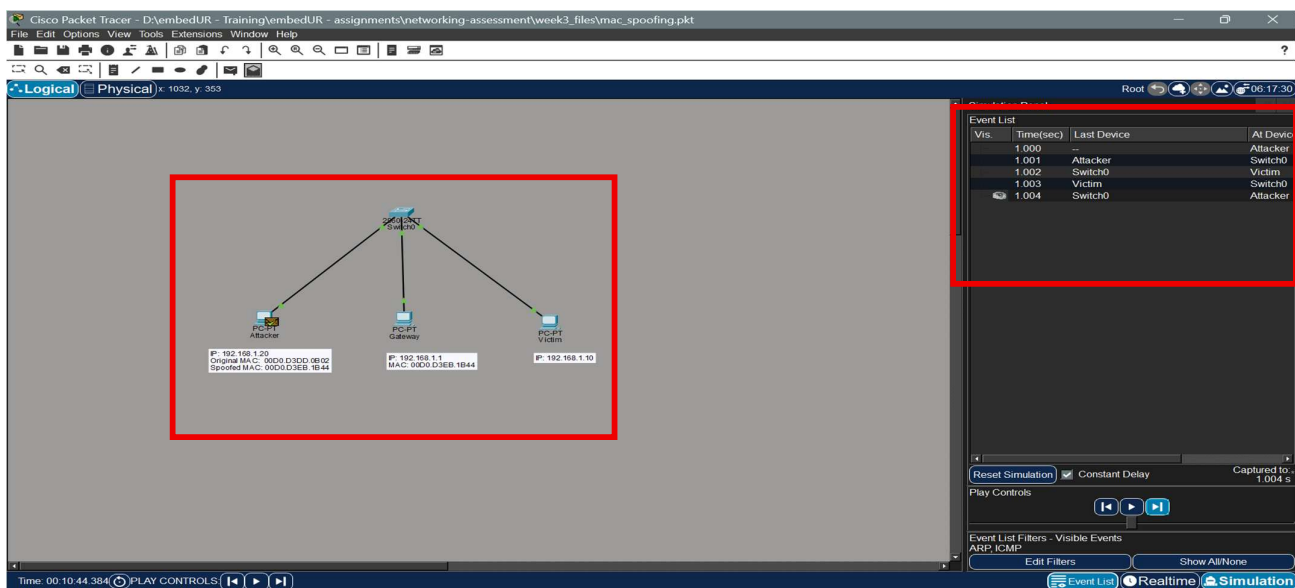
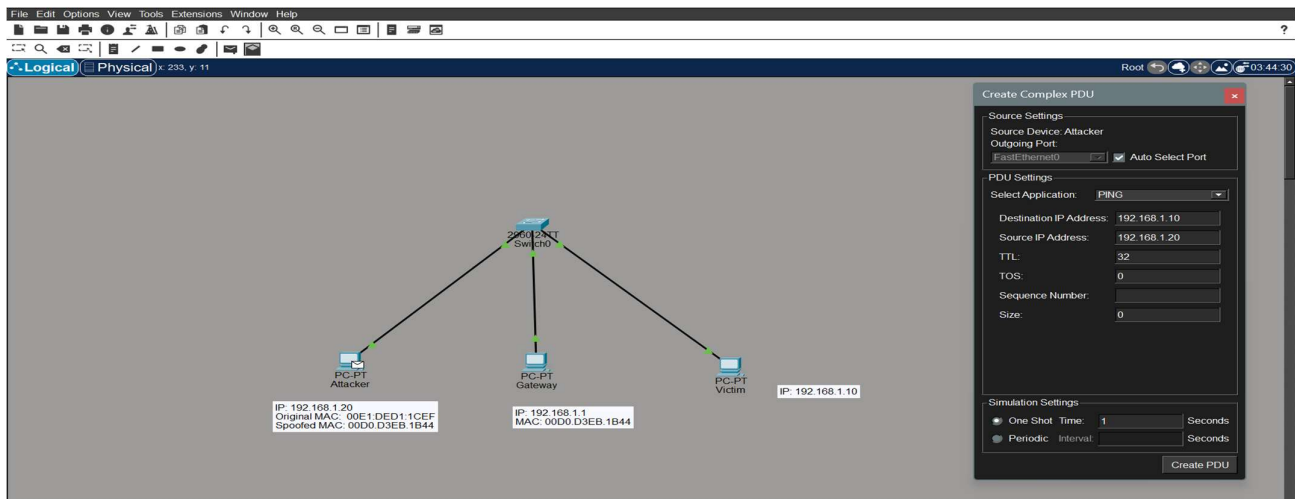
c. Establish Normal ARP Cache

- On the Victim PC, open the Command Prompt and ping the Gateway: ping 192.168.1.1
- Then type the following command on the Victim PC to verify the ARP table: arp -a
- Ensure it lists the correct MAC address of the Gateway: 00D0.D3EB.1B44

On this normal operation we get the following output



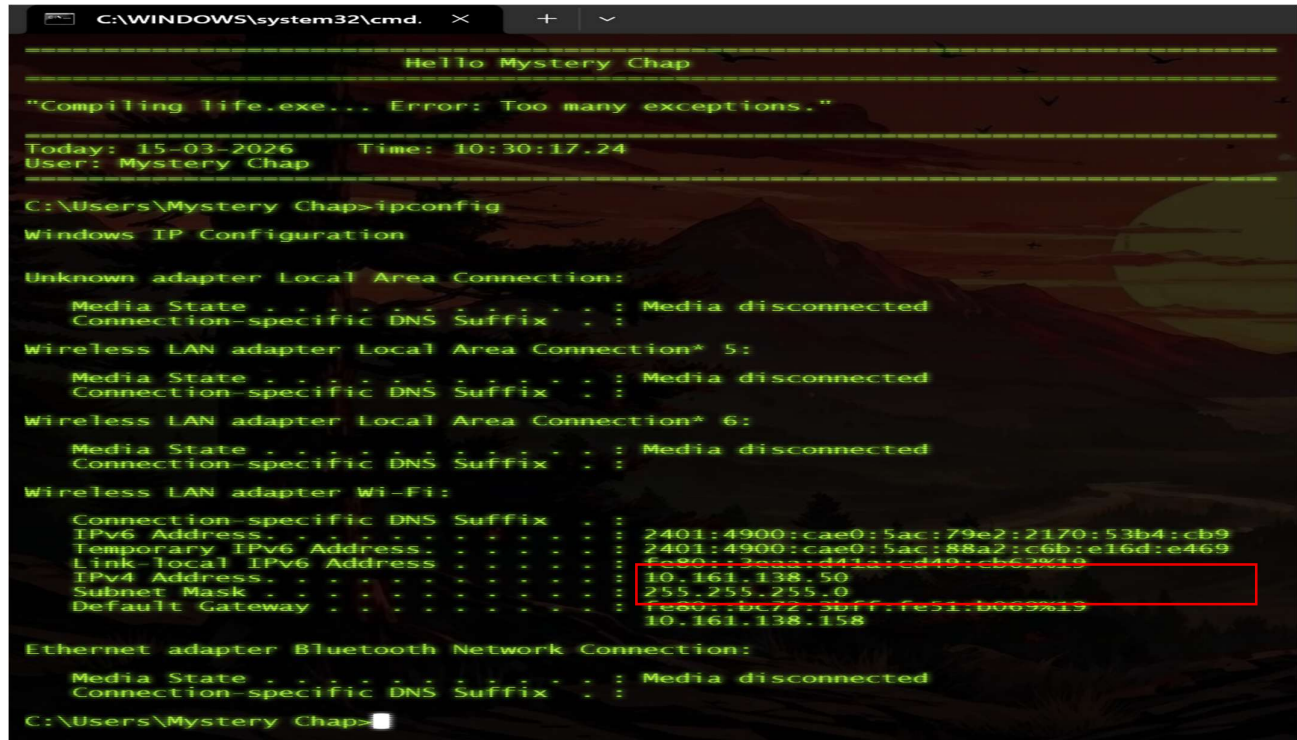
d. Now let us do ARP poisoning in the network. Create a Complex PDU packer after changing the MAC address of the attacker into the MAC of the Gateway. We can know the MAC Address of Gateway by pinging the gateway by this it gets stored in arp table. Now click on simulation. After simulation check on the Victim ARP cache it will be containing the attacker IP mapped to gateway MAC



Because of this until the cache gets updated the packets to gateway are sent to Attacker device. To prevent this we can enable port security (Sticky MAC) on switch to lock MAC address to specific port in switch, and use DAI (Dynamic ARP Inspection).

3. Configure static IP on client device (like PC or mobile phone) and verify connectivity using ping

a. To configure a static IP address, I must either have a static IP assigned to me or know the subnet of the network I am connected to and select an IP address that does not conflict with other devices. Since choosing a random IP address in a public network may cause an IP collision, I am using my mobile hotspot to ensure that I can safely assign a static IP within its subnet.



```

C:\WINDOWS\system32\cmd. x + ~
Hello Mystery Chap
"Compiling Life.exe... Error: Too many exceptions."
-----
Today: 15-03-2026 Time: 10:30:17.24
User: Mystery Chap
-----
C:\Users\Mystery Chap>ipconfig
Windows IP Configuration

Unknown adapter Local Area Connection:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 5:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 6:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

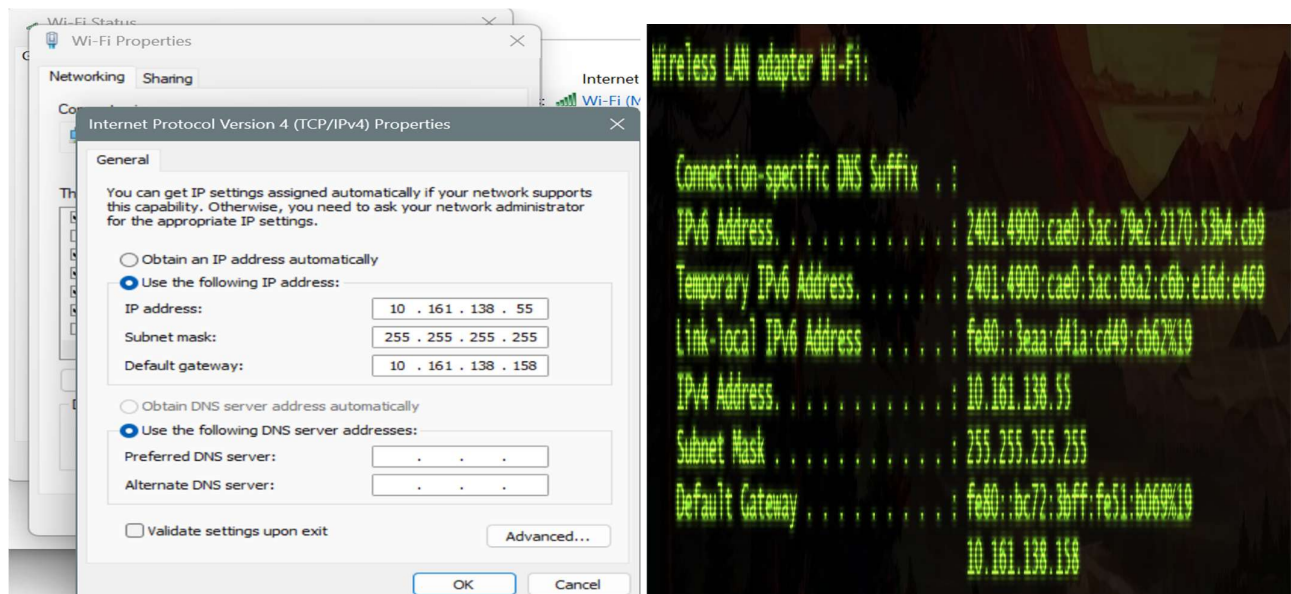
Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix . : 
    IPv6 Address. . . . . : 2401:4900:cae0:5ac:79e2:2170:53b4:cb9
    Temporary IPv6 Address. . . . . : 2401:4900:cae0:5ac:88a2:c6b:e16d:e469
    Link-local IPv6 Address . . . . . : fe80::3eaa:d41a:ed49:cb62%19
    IPv4 Address. . . . . : 10.161.138.50
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::bc72:3bff:fe51:b069%19
                                10.161.138.158

Ethernet adapter Bluetooth Network Connection:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

C:\Users\Mystery Chap>
    
```

The above picture shows DHCP assigned IP. Now we have to change the setting to assign a static IP

b. Go to Control Panel -> network and sharing -> Choose the network properties -> IPv4 properties and assign a static IP



c. Ping a domain to check it works

```

C:\WINDOWS\system32\cmd.
C:\Users\Mystery Chap>ping google.com

Pinging google.com [2404:6800:4007:838::200e] with 32 bytes of data:
Reply from 2404:6800:4007:838::200e: time=25ms
Reply from 2404:6800:4007:838::200e: time=150ms
Reply from 2404:6800:4007:838::200e: time=53ms
Reply from 2404:6800:4007:838::200e: time=45ms

Ping statistics for 2404:6800:4007:838::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 150ms, Average = 68ms

C:\Users\Mystery Chap>
  
```

4. Use Wireshark to capture DHCP Discover, Offer, Request and Acknowledge message and explain the process.

To initiate the DHCP process we need to run the following command to force ask the DHCP server a IP.

ipconfig /release – releases the current IP and the setting related to it.

ipconfig /renew – gets a new IP from the DHCP server. During the following command only we can capture the DHCP packets and DORA process takes place.

The screenshot displays two windows. On the left, a Windows Command Prompt shows the execution of `ipconfig /release` and `ipconfig /renew`. The output of `ipconfig /release` shows that the IP address has been released. The output of `ipconfig /renew` shows that a new IP address (10.161.138.50) has been assigned. On the right, a Wireshark capture of the network traffic shows the DHCP DORA process. The capture is filtered for DHCP packets. The following table summarizes the captured packets:

No.	Time	Source	Destination	Protoc	Length	Info
43	6.580	10.161.138.50	10.161.138.158	DHCP	342	DHCP Release - Transaction ID
460	12.70	0.0.0.0	255.255.255.2	DHCP	342	DHCP Discover - Transaction ID
461	12.72	10.161.138.158	10.161.138.50	DHCP	352	DHCP Offer - Transaction ID
462	12.72	0.0.0.0	255.255.255.2	DHCP	362	DHCP Request - Transaction ID
463	12.73	10.161.138.158	10.161.138.50	DHCP	352	DHCP ACK - Transaction ID

The Wireshark packet details for the selected packet (Frame 43) show the following structure:

- Frame 43: Packet, 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface 0
- Ethernet II, Src: Intel_55:61:7b (4c:49:6c:55:61:7b), Dst: 01:00:00:00:00:00
- Internet Protocol Version 4, Src: 10.161.138.50, Dst: 255.255.255.255
- User Datagram Protocol, Src Port: 68, Dst Port: 67
- Dynamic Host Configuration Protocol (Release)

The DHCP automatically assigns IP to the devices in the network using 4 steps. (DORA)

- a)Discovery: The client broadcasts a DHCPDISCOVER message to find available DHCP servers.
- b)Offer: DHCP servers receive the broadcast and respond with a DHCPOFFER, which includes an IP address, subnet mask, gateway, and lease duration.
- c)Request: The client receives the offer(s) and broadcasts a DHCPREQUEST, officially requesting the IP address from the chosen server.
- d)Acknowledgment: The server sends a DHCPACK to confirm the lease, allowing the client to start using the network.

The client also checks the network for duplicate IP. If duplicate IP exist then the client sends DHCPDECLINE after ACK also during renewal the client may ask for its previous held IP, if that is already assigned to another device then the server will give NAK (Negative ACK). Apart from this if a client and server are different subnets then it uses a DHCP relay. I have also captured the packet for DHCP connection. When a client releases an IP it sends DHCPRELEASE in the network.

The entire DHCP process uses the UDP protocol at the transport layer.

5. Given an IP address range of 192.168.1.0/24. Divide the network into 4 subnets

Task: Manually calculate the new subnet mask and the range of valid IP address for each subnet.

Assign IP addresses from these subnets to devices in CISCO Packet tracer and verify the connectivity using ping between them

Calculation

The given network is 192.168.1.0/24.

The CIDR /24 indicates that:

- Network bits: 24 bits
- Host bits: 8 bits

The total number of IP addresses available in this network is:

$2^8 = 256$ IP addresses

Since the task is to divide this network into 4 subnets, the number of subnet bits (n) must satisfy the condition: **$2^n \geq \text{number of required subnets}$**

For 4 subnets: $2^n \geq 4 \rightarrow n = 2$

Therefore, 2 bits must be borrowed from the host portion to create the required subnets.

Originally, the subnet mask for /24 was: 255.255.255.0

Binary: 11111111.11111111.11111111.00000000

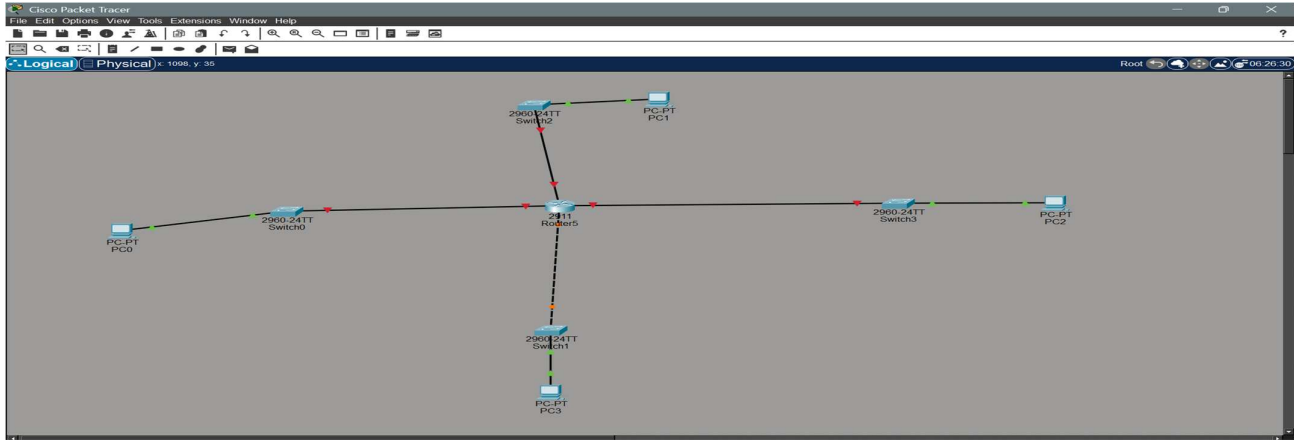
After borrowing 2 bits from the host portion, the new subnet mask becomes /26: 255.255.255.192

Binary: 11111111.11111111.11111111.11000000

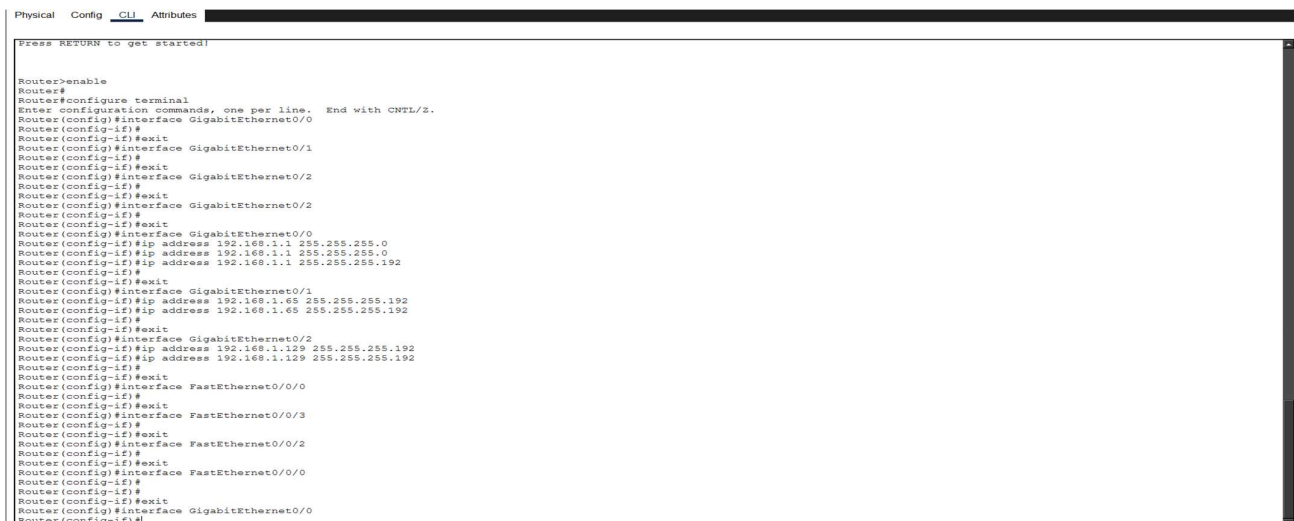
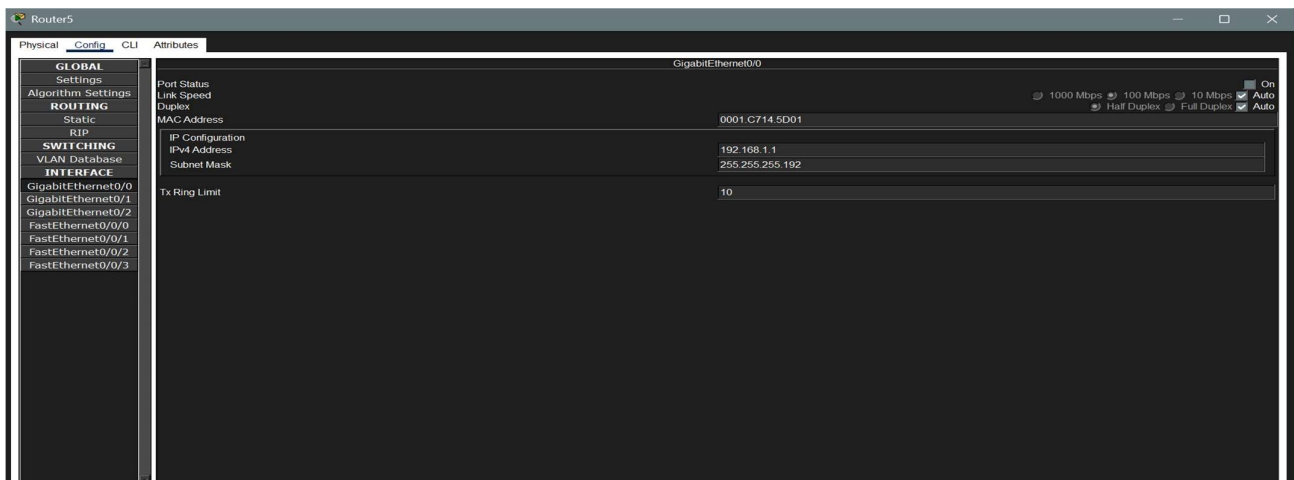
Thus, the network 192.168.1.0/24 is divided into 4 subnets using the new subnet mask /26 (255.255.255.192)

Building this network in CISCO Packet Tracer

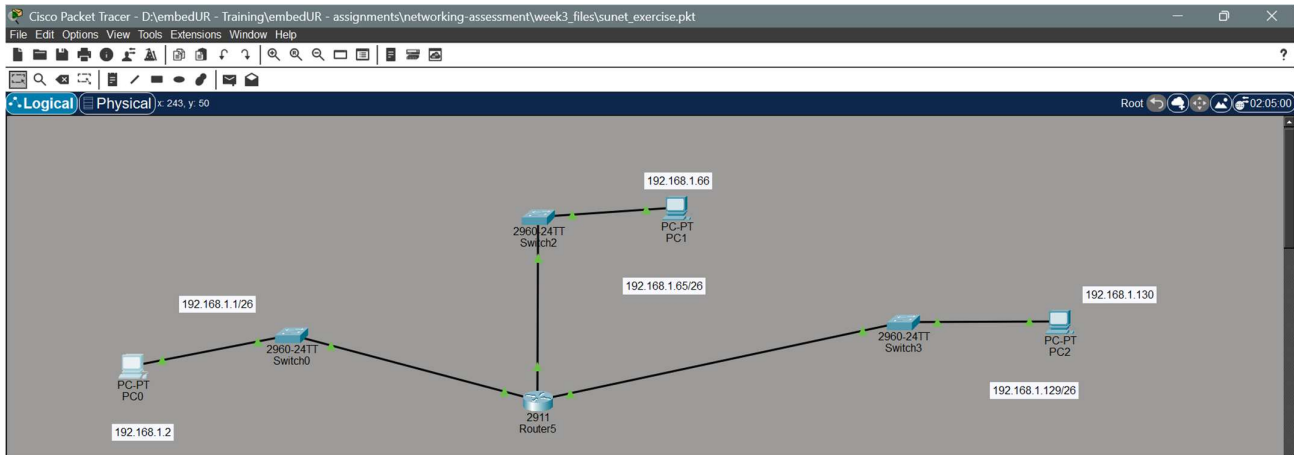
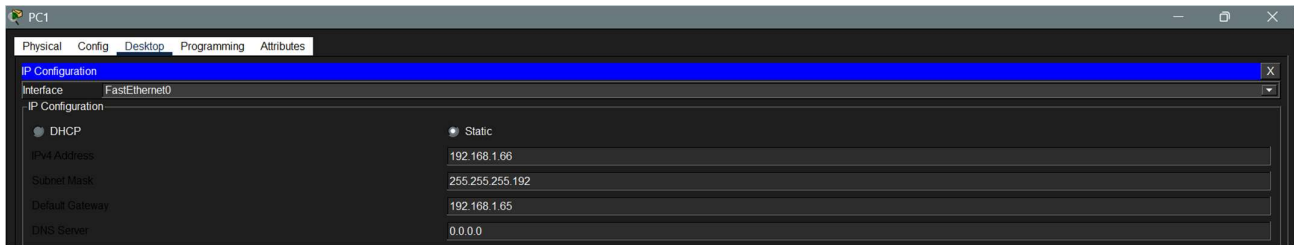
- a. Create a topology with a router and 4 switches and connect PC as required



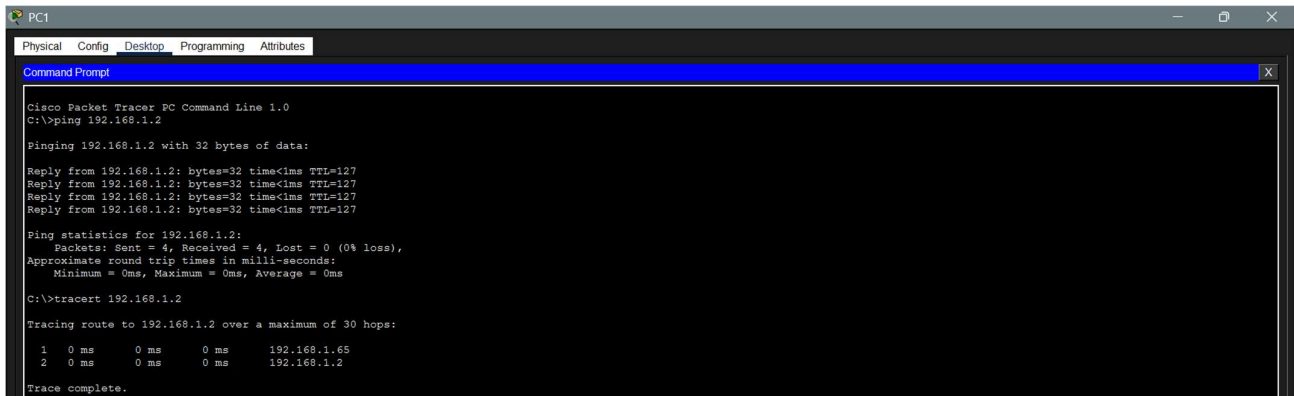
- b. Enter the IP address and subnet address in each port. Configure each port similarly



- c. Configure IP in each system according to its subnet along with the default gateway (IP address of the router port)



- d. Now ping one subnet from the other subnet



6. You are given 3 IP addresses: 10.1.1.1, 172.16.5.10 and 192.168.1.5.

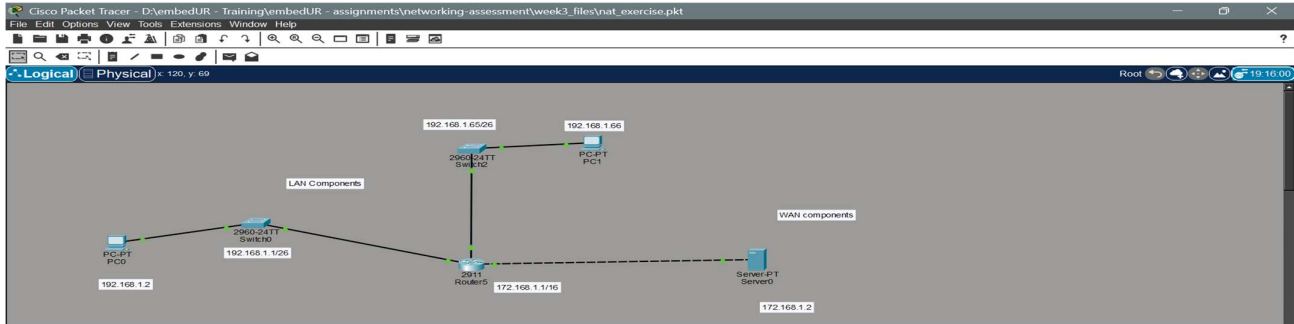
Task: Identify the class of each IP addresses (Class A, B and C). What is the default subnet mask for each class ?

Provide the range of IP address for Each Class

IP Address	Class	Subnet Mask	Range (first octet)
10.1.1.1	A	255.0.0.0	1 – 126
172.16.5.10	B	255.255.0.0	128 – 191
192.168.1.5	C	255.255.255.0	192 – 223

7. In the CISCO Packet Tracer, create a small network with multiple devices (e.g 2 PCs and a router). Use a private IP address (e.g 192.168.1.x) on the PC and configure the router to perform a network address translation to allow PC to access the internet.

a. Create a topology with 2 PC (can be 2 or more), a switch (optional), a router and a server (to act as WAN component)



b. Now configure the router component with the following commands

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Gig0/0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface Gig0/1
Router(config)#
Router(config)#interface GigabitEthernet0/0
Router(config-if)#interface Gig0/1
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#interface Gig0/2
Router(config-if)#ip nat outside
Router(config-if)#no shutdown

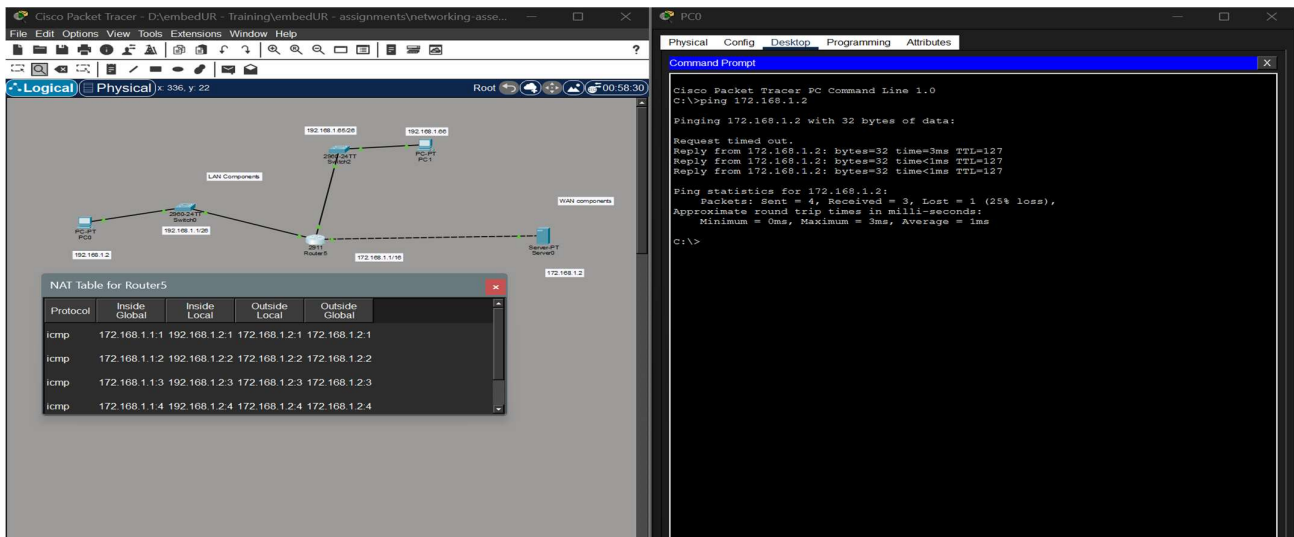
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

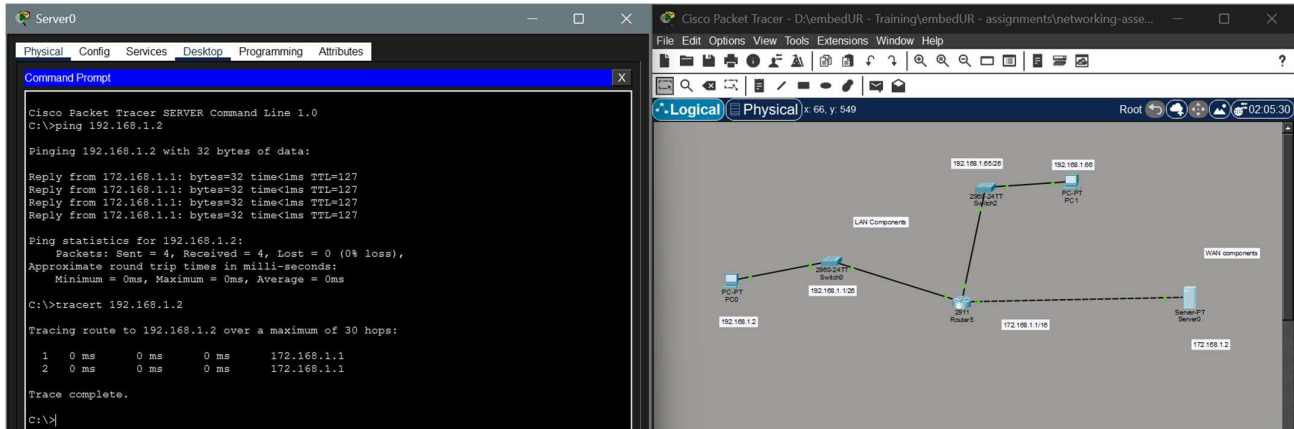
Router(config-if)#exit
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.63
Router(config)#access-list 1 permit 192.168.1.64 0.0.0.63
```

```
Router(config)#ip nat inside source list 1 interface Gig0/2 overload
```

c. Now ping the WAN component system from the LAN(private IP system) component



We can access the public component (WAN) from the private component but we can never access the private component from the public components.



It only reached the router from where the packets will be send based on the NAT table.